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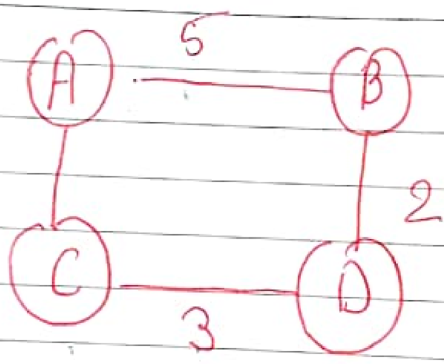
Minimum Spanning tree - A spanning tree can be defined as the subgraph of an undirected connected graph. A spanning tree is a subset of the graph that does not have a cycles, and it also cannot be disconnected.

A spanning tree consists of n vertices and n-1 edges. A complete undirected graph can have n^{n-2} no. of spanning tree where n is the no. of vertices in a graph.

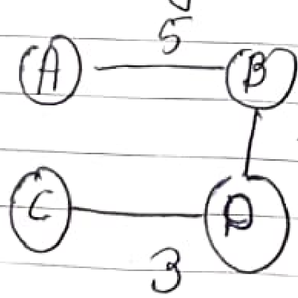
Ex -



Here vertex = 4

~~No of spanning tree = 4^{4-2} = 4^2 = 16~~

(I)



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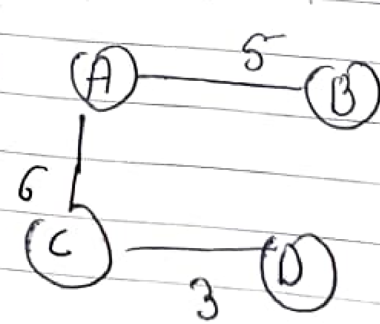
(II)



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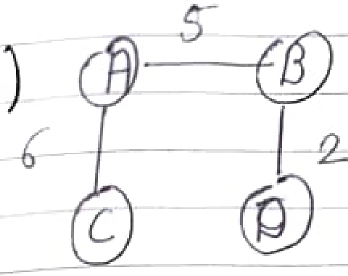
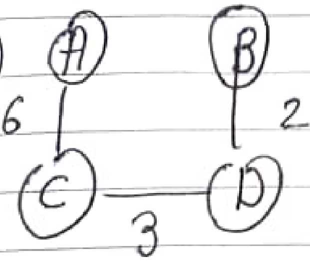
III)



IV)



13   11

A minimum tree selected from the above spanning trees is I). because it contain less weight.

Two algorithms are used for finding minimum spanning tree:-

(I) Kruskal algorithm   (II) Prim algorithm

Kruskal Algorithm is an algorithm that finds a minimum spanning tree for a connected weighted graph. It is edge based algorithm.

```
Mst_Kruskal (G, w)
    A ← φ
    for each vertex v ∈ V[G]
        do make-set (v)
    sort the edges of E into non-decreasing order by
    weight w
    for each edge (u,v) ∈ E, taken in non-decreasing
    order by weight
        do if Find_set(u) ≠ Find_set (v)
            then A ← A ∪ {(u,v)}
                Union (u,v)
        return A.
```

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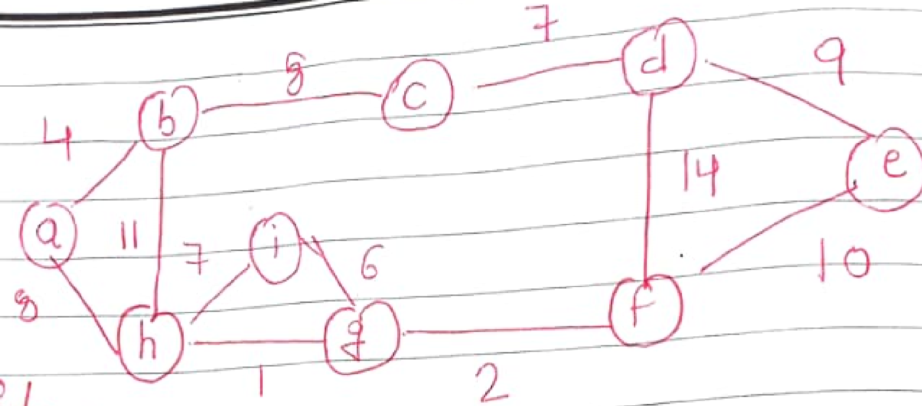

Sort Edges

(h, g)	1
(g, f)	2
(a, b)	4
(i, g)	6
(h, i)	7
(c, d)	7
(b, c)	8
(a, h)	8
(d, e)	9
(e, f)	10
(b, h)	11
(d, f)	14

Intially $A = \phi$

Make-Set(v)

$\{a\}$

$\{b\}$

$\{h\}$

$\{i\}$

$\{g\}$

$\{c\}$

$\{d\}$

$\{f\}$

$\{e\}$

So first take edge (h, g)

$A = \{h, g\}$

Union(u,v) = $\{h, g\}$

then edge (g, f)

$A = \{h, g, f\}$

Union(u,v) = $\{h, g, f\}$

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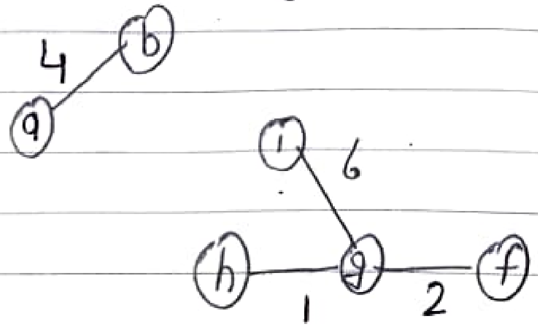
Then (a,b) and (i,g) edge are considered

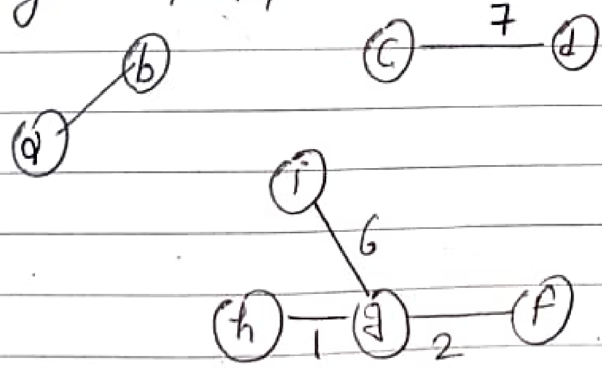


A = { h, g, f, a, b, i, g }

Union(u,v) = { h, g, f, a, b, i, g }

Now edge (h,i). Both (~~h, i~~) h and i are in same set, thus create a cycle. So this edge discarded.

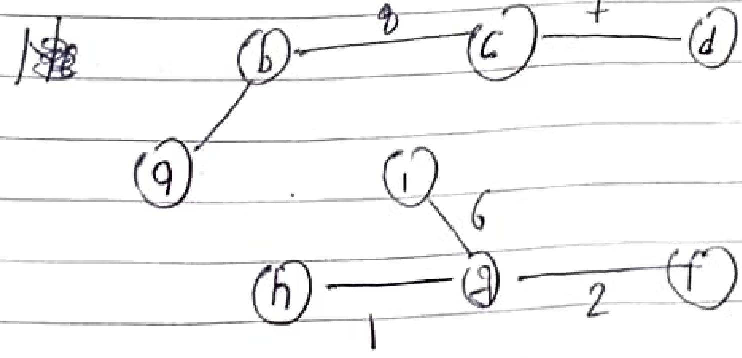
Next edge (c,d),



A = { h, g, f, a, b, i, g, c, d }

Union(u,v) = { h, g, f, a, b, i, g, c, d }

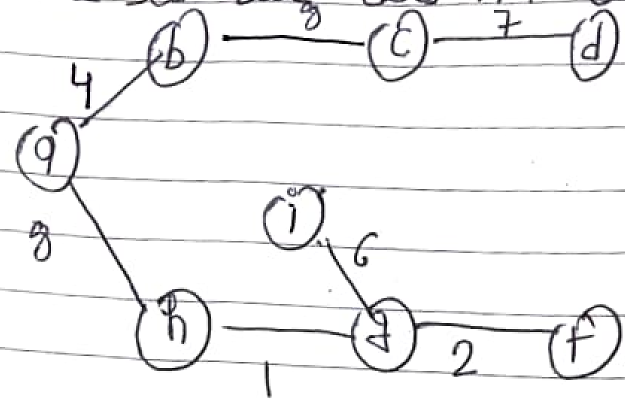
Next edge (b,c), both b and c are in same set



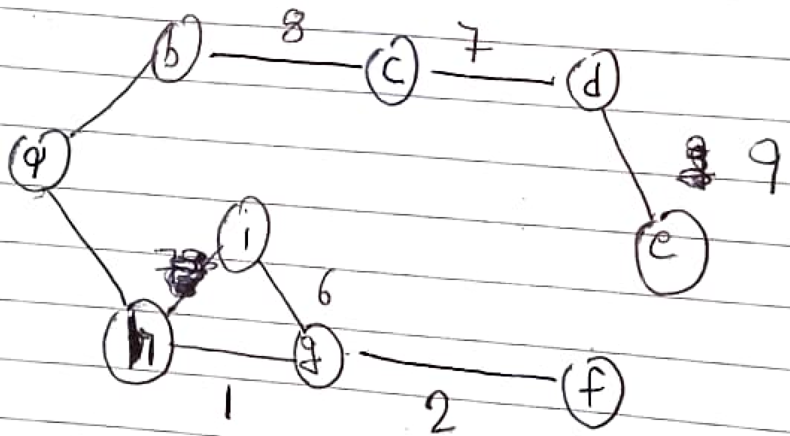
A = { h, g, f, a, b, i, c, d }

Union(u,v) = { h, g, f, a, b, i, c, d }

Next edge (a, h), both a and h are belong to same set but does not create a cycle.



Next edge (d, e)



$A = \{h, g, f, a, b, i, c, d, e\}$

$Union(u,v) = \{h, g, f, a, b, i, c, d, e\}$

Next edge (e, f)

Both e and f belong to same set and it will create a cycle.

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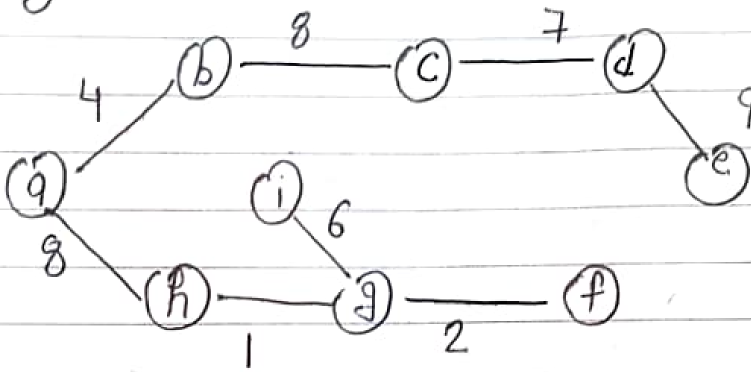
Next edge (b, h)

Both b and h belong to same Set and it will create a cycle.

Next edge (d, F)

Both edge d and f belong to same set and it will create a cycle.



Time Complexity $= O(E \log E)$

Prim's algorithm - The idea of prims algorithm is similar to that of Dijkstra's algorithm for finding shortest path in a given graph.

MST - PRIM (G, w, r)
for each $u \in V[G]$
do Key $[u] \leftarrow \infty$
$\pi[u] \leftarrow$ Nil
Key $[r] \leftarrow 0$
$Q \leftarrow V[G]$
While $Q \neq \phi$
do $u \leftarrow$ Extract_Min(Q)
for each $v \in Adj[u]$
do if $v \in Q$ and $w(u, v) <$ Key$[v]$

$- O(V)$

$- O(V \log V)$

V^2 for complete graph

$O(V) + O(V \log V) + O(V^2 \log V)$

$= O(E \log V)$ [for complete graph]

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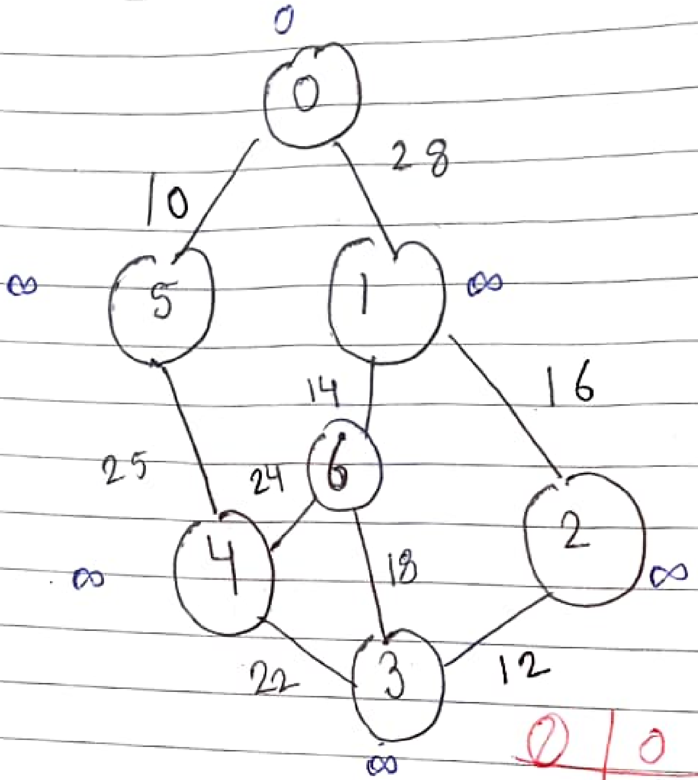
Then $\pi[v] \leftarrow u$

$Key[v] \leftarrow w(u,v)$



Q	0	1	2	3	4	5	6
	$\boxed{0}$	∞	∞	∞	∞	∞	∞

In Prim's algorithm, first we intialize the priority Q to contain all the vertices and the Key of each vertex to ∞ except from the root, whose Key is set to 0.

Q	0	1	2	3	4	5
		28	∞	∞	∞	$\boxed{10}$

$Key[5] = \infty$

$w(0,5) = 10$

$\pi[5] = 0$ and $Key[5] = 10$

$Key[1] = \infty$

$w(0,1) = 28$ i.e. $w(u,v) < Key[v]$

$\pi[1] = 0$ and $Key[1] = 28$

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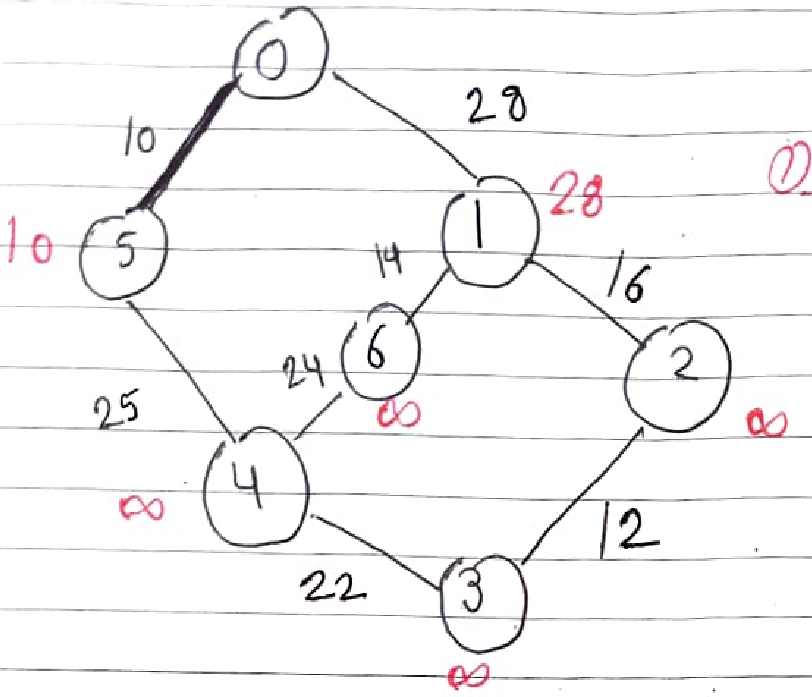
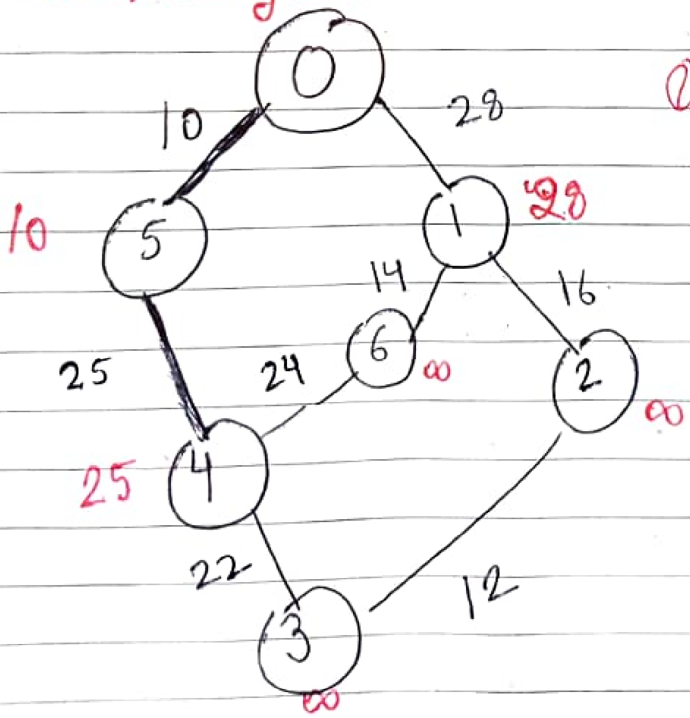


0	1	2	3	4	6
	∞	∞	∞	$\boxed{25}$	∞

How, by Extract_min(0) remove 5 because
Key[5] = 10
Adj[5] = 4

w(u,v) < Key[v] then Key[4] = 25
w(5,4) < Key[4]



0	1	2	3	4	6
	28	∞	∞	$\boxed{25}$	∞

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How by Extract_min(Q) remove 4 because Key[4] = minimum

Key[4] = 25   u = 4.

Q	1	2	3	6
	28	∞	$\boxed{22}$	24

Adj[4] = {6,3}

w(u,v) < Key[V]

then Key[6] = 24

Key[3] = 22 is minimum.

How by Extract_min(Q) remove 3

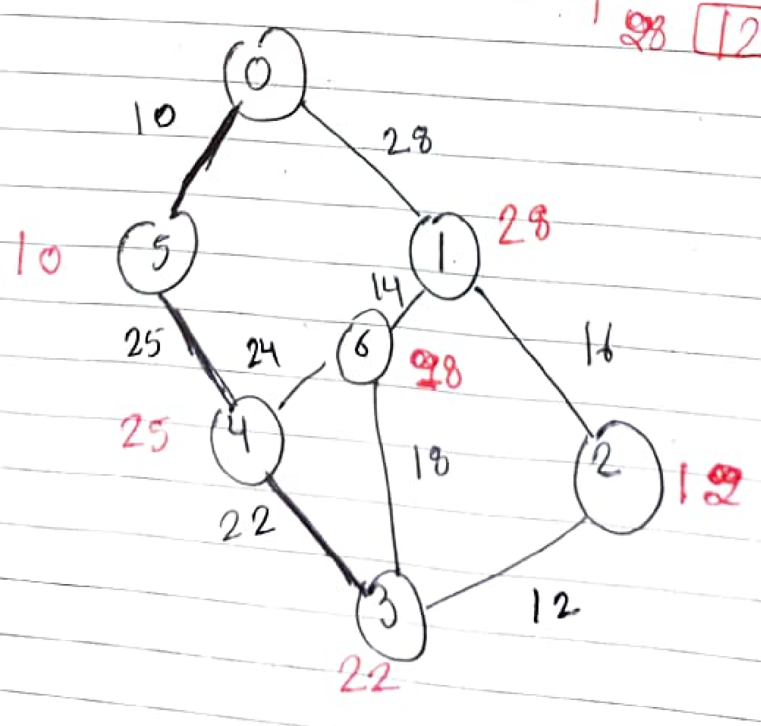
Key[3] = 22

Adj[3] = {4,6,2}

$4 \notin Q$ Key[6] = 18

Key[2] = 12

Q	1	2	6
	28	$\boxed{12}$	18



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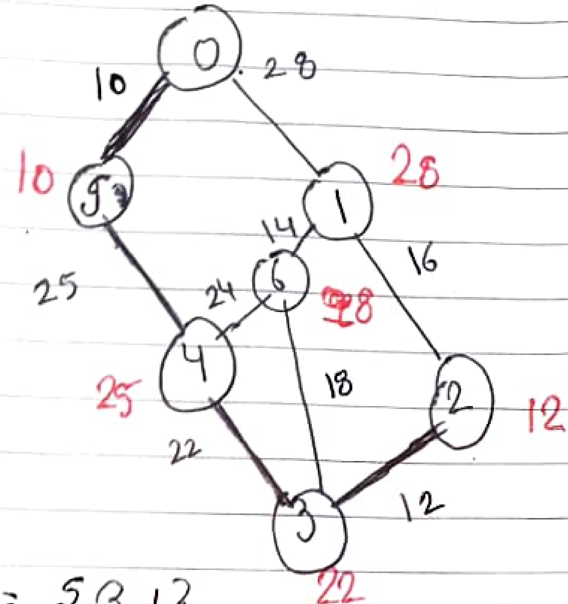
How by Extract_min(Q) Remove 2, because Key[2] is minimum



Adj[2] = {3,1}

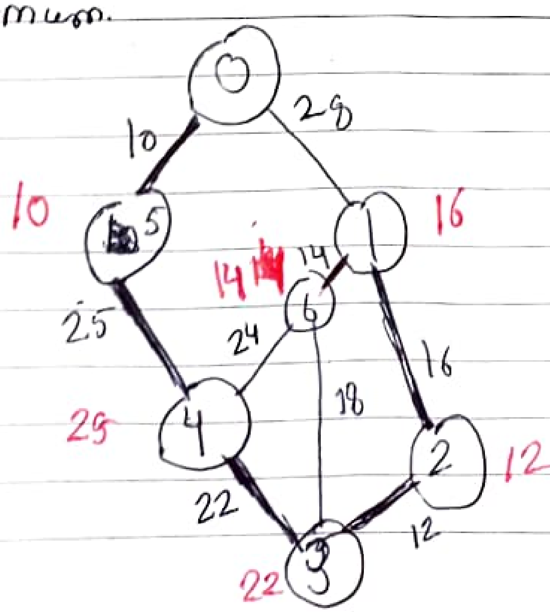
~~$3 \neq Q$~~

$3 \neq Q$

Q	1	6
	$\boxed{16}$	28

Key[1] = 16

How by Extract_min(Q) remove 1, because Key[1] is minimum.



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Adj [1] = {0, 6, 2}

2, 0 ≠ Q

0 | 6
$\boxed{24}$

Key [6] = 14

By Extract_min(Q) remove vertex 6.

Time Complexity of $O(V^2)$ if the input graph is represented using adjacency list. $O(E \log V)$ with binary heap.

Unit - 4

Naive String Matching Algorithm

Naive_String_Matcher (T, P)
  n ← length [T]
  m ← length [P]
  for s ← 0 to n-m
    do if P[1 .. m] = T [s+1 . . . s+m]
      then print " Pattern occurs with shift s

Naive_string_Matcher (T, P)
1. n ← length [T]
2. m ← length [P]
3. for s ← 0 to n-m do
4.   j ← 1
5. while j ≤ m and T[s+j] = p [j] do
6   j ← j+1
7. If j > m then
8   return valid shift s
9. return no valid shift exist